



Yiming Bao

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EDUCATION

• Undergraduate

Department of Computer Science and Technology, Tsinghua University
(advisor: Huazhe Xu)
GPA: 3.8/4.0

2023 - Present
Beijing, China

• Selected Major Courses

CS: Programming and Training (4.0), Data Structure (4.0), Software Engineering (4.0), High Performance Computing (4.0), Digital Logic Circuit (4.0), Media Computing (4.0), Computer Network (4.0)

AI/ML: Introduction to Artificial Intelligence, Artificial Neural Network (4.0)

Math/Physics: Probability and Statistics (4.0), Linear Algebra (4.0), Calculus (4.0), Complex Analysis (4.0), Physics(I)(III) (4.0)

PUBLICATIONS & PREPRINTS

• DexImit: Learning Bimanual Dexterous Manipulation from Monocular Human Videos

RSS 2026 (In Submission) [Project] [Paper]

Juncheng Mu*, Sizhe Yang*, Yiming Bao, Hojin Bae, Tianming Wei, Linning Xu, Boyi Li[†], Huazhe Xu[†], Jiangmiao Pang[†]

• UniDex: A Robot Foundation Suite for Universal Dexterous Hand Control from Egocentric Human Videos

CVPR 2026 [Paper]

Gu Zhang*, Qicheng Xu*, [...], Yiming Bao*, [...], Yang Gao[†], Hao Zhao[†], Hang Zhao[†], Huazhe Xu[†] (19 authors)

RESEARCH

• Research Interest

My long-term research goal is to develop robots with human-level versatility in perception and reasoning, ultimately enhancing everyday life at scale.

My current focus lies at the intersection of robotics and machine learning, particularly in building general-purpose manipulation systems that can robustly operate in unstructured real-world environments.

• Experience

– DexImit: Learning Bimanual Dexterous Manipulation from Monocular Human Videos

Nov 2025 - Jan 2026

- * Try to deal with **data scarcity** that limits the generalizations of bimanual dexterous manipulation using **generated or in-the-wild human videos**
- * A simple and effective framework automatically converts **monocular human videos** into **physically plausible robot demonstrations** via 4D reconstruction, action-centric scheduling, force-closure grasp synthesis, and large-scale augmentation
- * enabling zero-shot real-world deployment **without any robot data**

– UniDex: A Robot Foundation Suite for Universal Dexterous Hand Control from Egocentric Human Videos

Jun 2025 – Nov 2025

- * Try to address **data scarcity and embodiment heterogeneity** in dexterous manipulation by leveraging large-scale **egocentric human videos** to build a robot-centric foundation dataset
- * Construct **UniDex-Dataset** (10M frames, 50K trajectories, 8 hands) and propose a unified **Function-Actuator-Aligned Space (FAAS)** with a pretrained **3D VLA policy** for cross-hand dexterous control
- * Enabling strong **zero-shot spatial, object, and cross-hand generalization**, significantly outperforming prior VLA baselines in real-world tool-use tasks

SKILLS

- **Language:** TOEFL (110/120, 5.5/6.0), CET-4, CET-6
- **Programming and Scripting Languages:** Python, C/C++, Rust, Java, JavaScript/TypeScript, CUDA, SystemVerilog
- **Professional Software:** Pytorch, ROS, Sapien, IsaacGym, Polymetis
- **Hardware and Robots:** Franka Panda, Inspire Hand, Wuji Hand, X Hand, Robotiq Gripper, Apple Vision Pro, Oculus Quest, Realsense Camera, NUC